

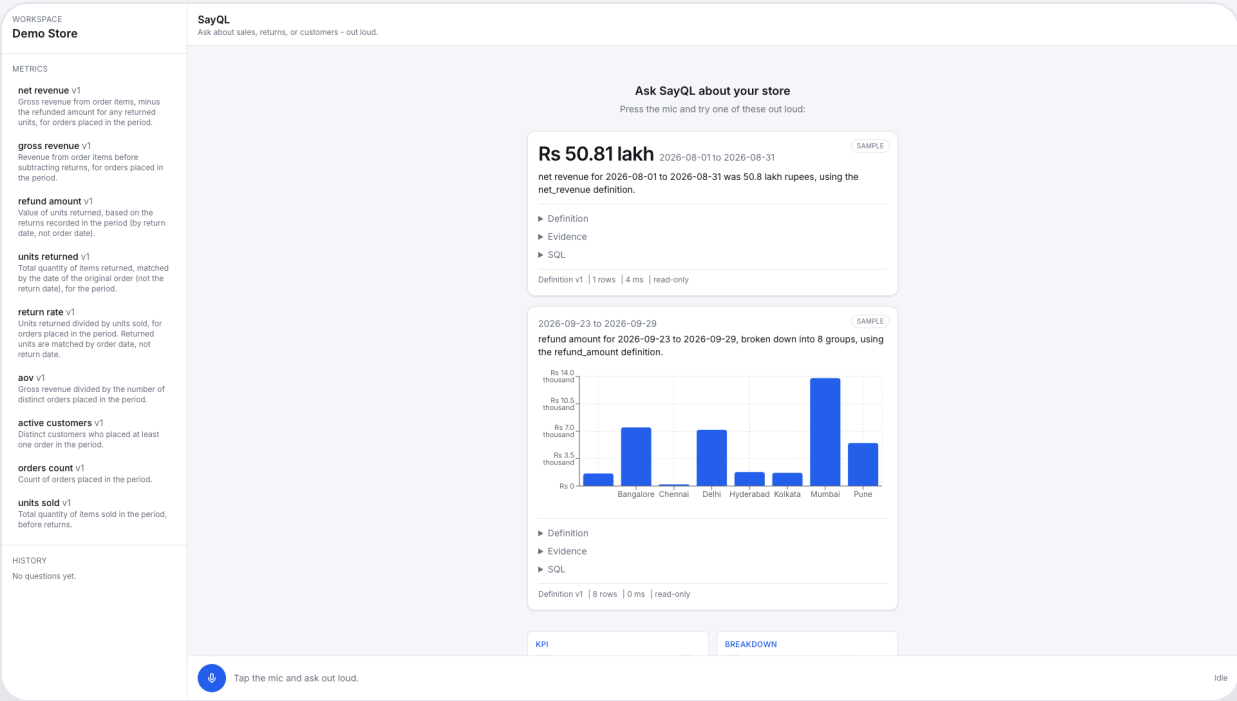
SayQL

A voice analyst that shows its definition.

Built for small D2C store owners who have data, but no analyst.

Live: sayql.onrender.com

Repo: github.com/23f1002033/sayql



PROBLEM

The same question can mean two different numbers.

Dashboards and plain voice-to-SQL just answer. They do not know a business term has more than one valid definition - so they guess, silently.

"Revenue", August 2026 - gross

Rs 53.82 lakh

"Revenue", August 2026 - net

Rs 50.81 lakh

Same word, same month

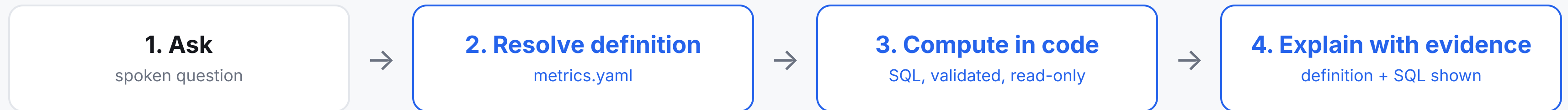
Rs 3.01 lakh gap

Nobody can trust a number they cannot see the definition of.

SOLUTION

Ask in words. Answer in code you can check.

SayQL never lets the model guess a business term or compute a number itself. It resolves a definition, computes in deterministic code, and explains with evidence.



Every number states its own definition.

Rs 50.81 lakh

2026-08-01 to 2026-08-31

net revenue for 2026-08-01 to 2026-08-31 was 50.8 lakh rupees, using the net_revenue definition.

▼ Definition

net revenue (v1)

Gross revenue from order items, minus the refunded amount for any returned units, for orders placed in the period.

▼ Evidence

resolved metric: net_revenue (v1)

period: 2026-08-01 to 2026-08-31

1 row(s) in 19 ms

▼ SQL

SELECT SUM(oi.quantity * oi.unit_price) - COALESCE(SUM(r.quantity * oi.unit_price), 0) AS value FROM order_items AS oi JOIN orders AS o ON o.order_id = oi.order_id JOIN products AS p ON p.product_id = oi.product_id LEFT JOIN returns AS r ON r.order_item_id = oi.order_item_id WHERE o.workspace_id = :workspace_id AND o.order_date BETWEEN :start_date AND :end_date LIMIT 200

Copy

Definition v1 | 1 rows | 19 ms | read-only

"WHAT WAS NET REVENUE LAST MONTH?"

Net revenue for August 2026 was Rs 50.81 lakh, spoken as "50.8 lakh rupees" - using the net_revenue v1 definition. The exact SQL that produced it is one tap away.

- The definition is versioned - v1, not just a name
- The SQL behind it is read-only, every time

It asks instead of guessing.

'revenue' could mean gross_revenue or net_revenue. Which one do you want?

Say: gross revenue

Say: net revenue

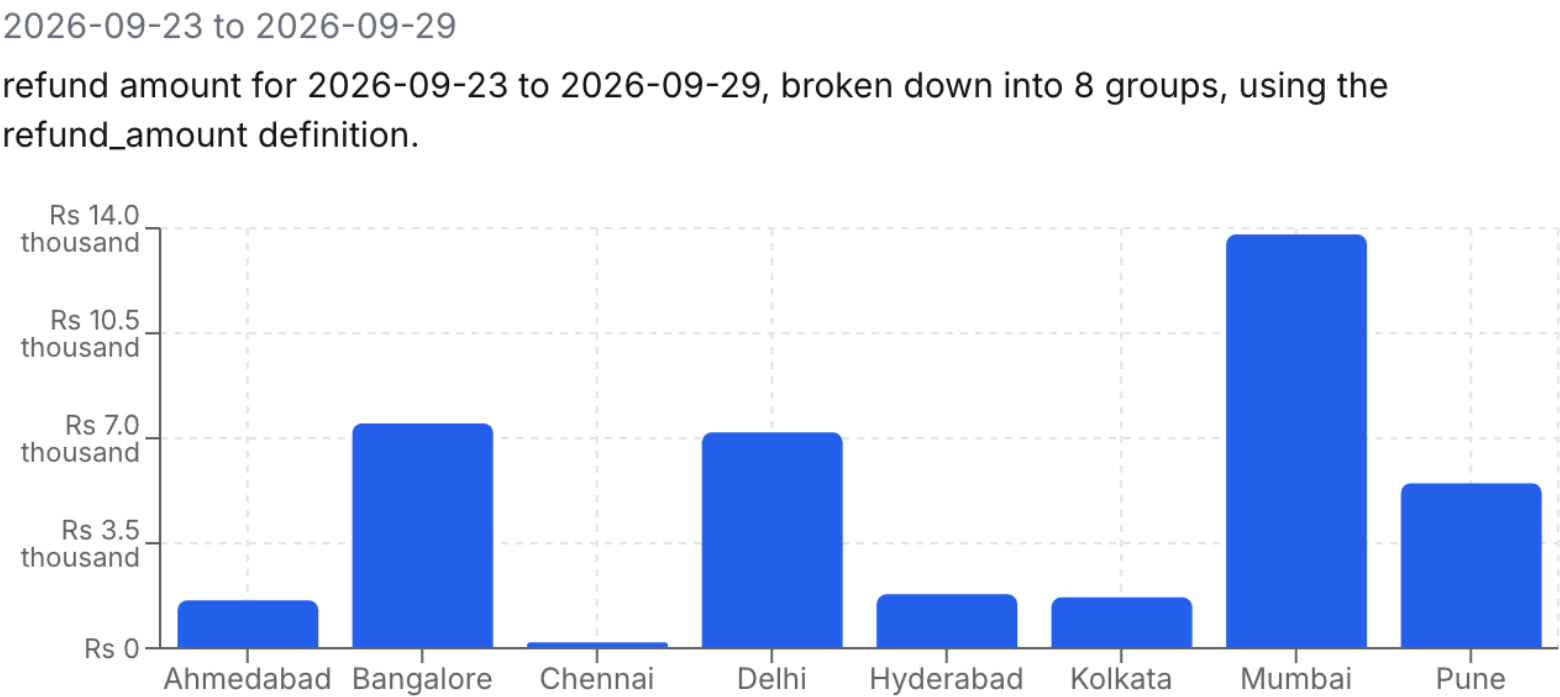
The Voice Agent API has no text-input channel, so these are reminders of the wording, not buttons - say your choice out loud.

"WHAT WAS OUR REVENUE LAST MONTH?"

"Revenue" is deliberately undefined. SayQL asks one short question back instead of picking a side. The options on screen are reminders of the wording, not buttons - there is no text-turn message in the Voice Agent API, so the answer has to be spoken.

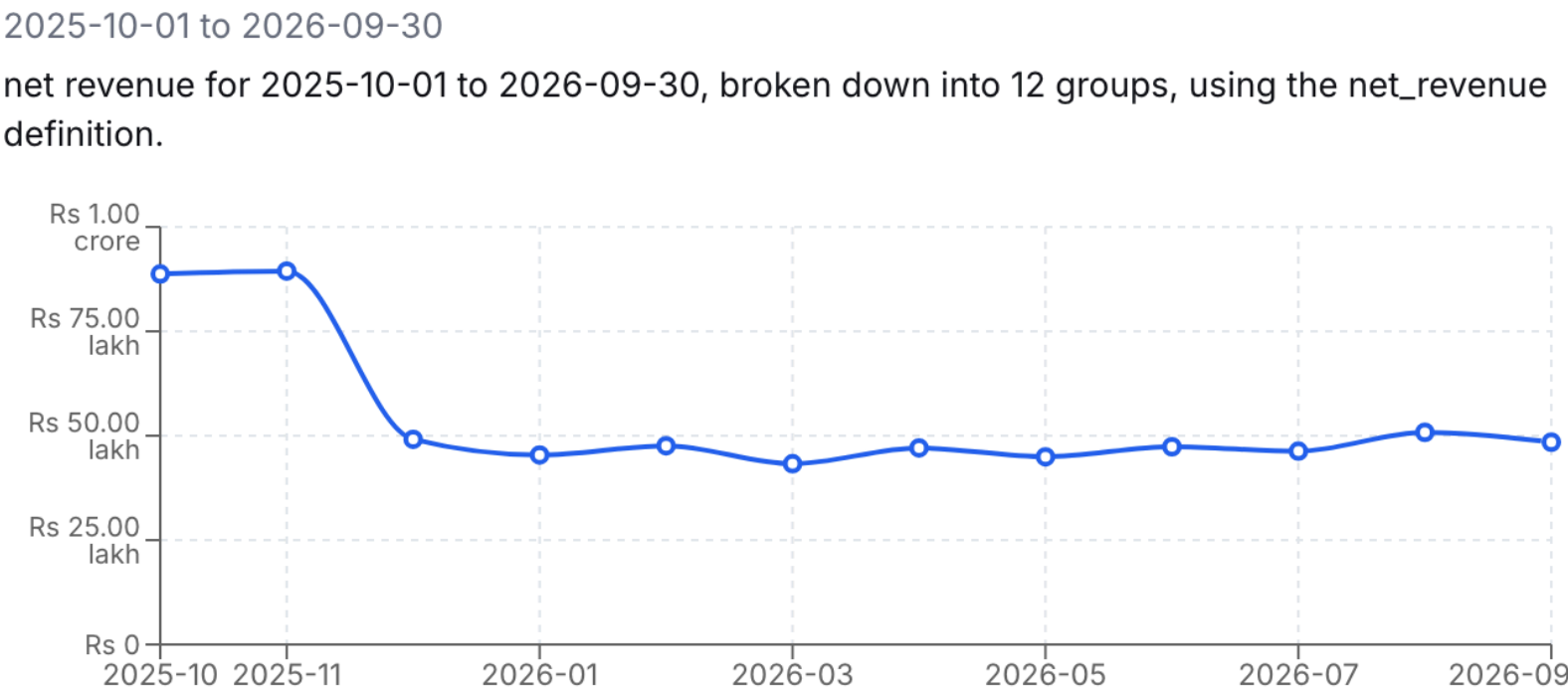
One dictionary, every shape of answer.

The same metric definitions drive a bar chart, a trend line, or a plain number - no new SQL written for each shape.



- Definition
- Evidence
- SQL

Definition v1 | 8 rows | 0 ms | read-only



- Definition
- Evidence
- SQL

Definition v1 | 12 rows | 47 ms | read-only

"What were returns last week by city?"

"Show me net revenue by month for the last year."

It explains, and says what it does not know.

Rs 5.0 thousand

Rs 40.0 thousand

up Rs 35.0 thousand

refund amount went from 5 thousand rupees to 40 thousand rupees (up 35 thousand rupees). Mumbai was the top contributor at 64.3 percent of the change. units sold also rose 533 percent, so part of the increase is higher volume, not only a higher return rate. This shows correlation, not proven cause.

units sold also rose 533 percent, so part of the increase is higher volume, not only a higher return rate.

Mumbai

Hyderabad

Ahmedabad

Rs 0

Rs 6.0 thousand

Rs 12.0 thousand

Rs 18.0 thousand

Rs 24.0 thousand

► Definition

► Evidence

Definition v1

13.33%

16.84%

up 3.5 percentage points

return rate went from 13.33 percent to 16.84 percent (up 3.5 percentage points). both moved sharply: units returned went from 2 to 16, units sold from 15 to 95. the earlier period had only 2 units returned, so the rate is noisy. This shows correlation, not proven cause.

Note: the earlier period had only 2 units returned, so the rate is noisy.

units returned

units sold

before

2 units

before

15 units

now

16 units

now

95 units

both moved sharply: units returned went from 2 to 16, units sold from 15 to 95.

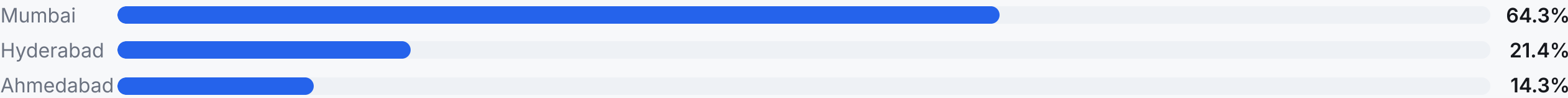
► Definition

► Evidence

Definition v1

ADDITIVE: REFUND AMOUNT

Refund amount for Wireless Earbuds Pro went from Rs 5.0 thousand to Rs 40.0 thousand.

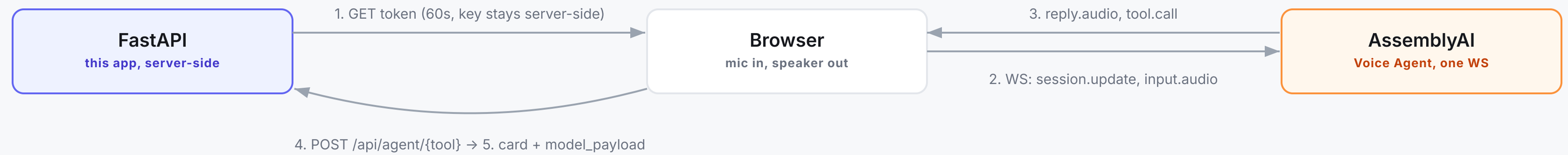


It also refuses to overclaim: units sold rose 533% too, so part of the jump is volume, not only a higher return rate.

RATIO: RETURN RATE

Return rate rose from 13.33% to 16.84% (up 3.5 percentage points). SayQL flags it: the earlier period had only 2 units returned, so the rate is noisy - a low_base warning, not a hidden number.

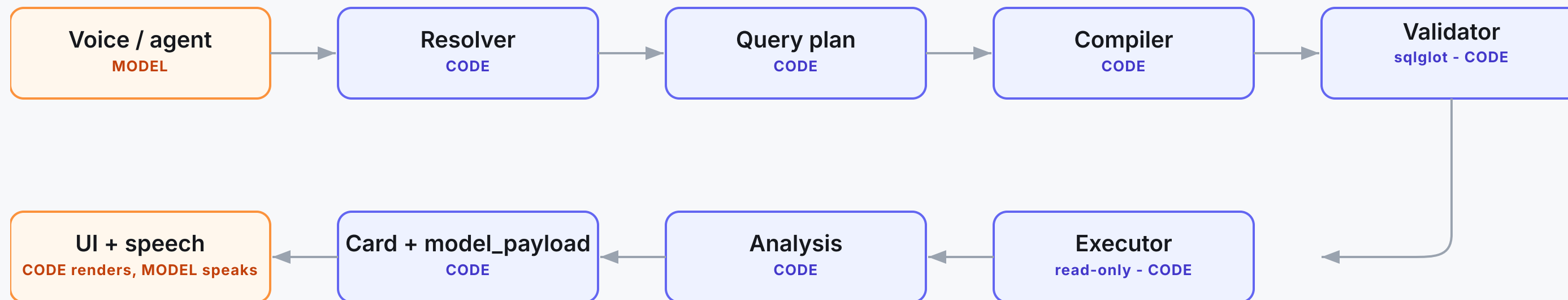
One WebSocket does speech, language, and tool calls.



- Server mints a 60 second token per session - the AssemblyAI API key never reaches the browser.
- Barge-in resets the playback schedule the instant the caller talks over a reply.
- tool.result is only sent after reply.done, per the Voice Agent API's own tool-calling rule - not immediately on tool.call.

ARCHITECTURE

Two boxes touch the model. The rest is code you can read.



Numbers are computed once, in Python, and formatted once, in app/format.py. The model only ever reads a pre-formatted spoken string back.

The bug that taught us not to trust the model with numbers.

BEFORE

Gross revenue was Rs 53.82 lakh. The agent spoke "five lakh thirty eight thousand rupees" - about 10 times too small. The model was converting the raw number itself.

FIX

Raw numbers were removed from everything the model sees. It now only reads pre-formatted spoken strings from model_payload, and is told never to compute or round a number itself.

100

automated tests passing

20233

orders in the seeded demo data

1501

returns tracked in the demo data

Accuracy eval: not yet run. evals/results.md does not exist yet - scripts/build_slide_numbers.py will pick it up once it does.

Ask works today. Analyze, Explain, Detect, Alert, Act are the roadmap.

Today LIVE

- Ask: KPI, breakdown, trend, and why-with-evidence answers, spoken and on screen
- A versioned metric dictionary, not freehand SQL
- CSV ingestion into the same schema
- Single-round explain_change: one dimension, one comparison

Roadmap NOT BUILT

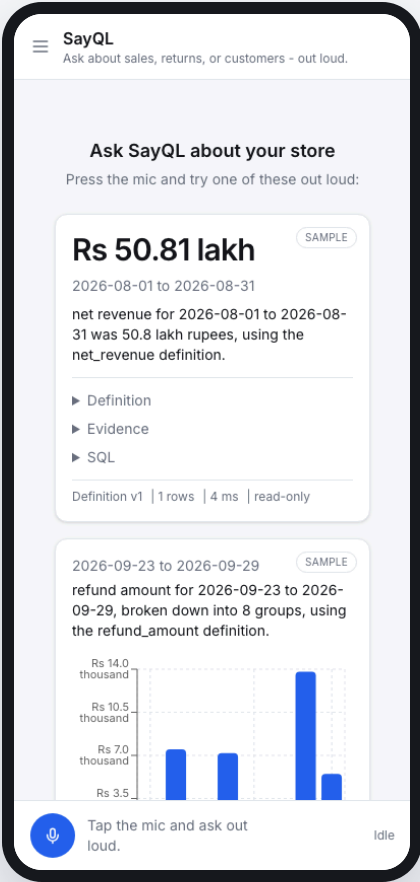
- Analyze: compare several dimensions at once, chain follow-up questions
- Explain: propose the next question instead of waiting for it
- Detect: a scheduled job that flags a change worth asking about
- Alert and Act: notify outside the app, then trigger a real action
- Shopify / WooCommerce connectors, auth and per-tenant isolation

TRY IT

Say four sentences. See the definition behind every one.



sayql.onrender.com



1. "What was net revenue last month?"
2. "What were returns last week by city?"
3. "What was our revenue last month?"
4. "Why did returns go up for the Wireless Earbuds Pro this month?"

Use headphones or a quiet room - open speakers can make the agent hear itself. Demo is rate-limited to 8 voice sessions per connection per 10 minutes, 300 total per day.